Homework - IV

Statistics III : Multivariate Data and Regression

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Question 1

Classify the following variables for a BSDS student as ordinal, nominal and continuous.

(a) Location (Bangalore/Delhi/Kolkata)

(b) Frequency of physical exercise (Never/Rarely/Often/Regularly)

(c) Percentage marks in first year

Solution 1

(a) Location is **nominal**. The categories are only labels and there is no natural ordering among Bangalore, Delhi and Kolkata.

(b) Frequency of physical exercise is **ordinal**. The categories are ordered as Never $<$ Rarely $<$ Often $<$ Regularly, but the gaps between consecutive categories are not meaningful (and need not be equal).

(c) Percentage marks is **continuous**. It takes values in the interval $[0, 100]$ and differences and ratios are meaningful.

Question 2

$(X_1, X_2, X_3) \sim \text{Multinomial}_3(n, (p_1, p_2, p_3))$ where $p_1 + p_2 + p_3 = 1$. Let $\hat{p}_i = X_i/n$ for $i = 1, 2$. The marginal distribution of X_i is $\text{Bin}(n, p_i)$, $i = 1, 2$.

(a) Find the expectation and variance of $\hat{p}_i$, $i = 1, 2$.

(b) Write down the joint distribution of (X_1, X_2).

(c) Using this show that the covariance of $(\hat{p}_1, \hat{p}_2)$ is $-p_1 p_2/n$.

(d) Using the previous two parts, find the variance of $(\hat{p}_1 - \hat{p}_2) = \sigma^2$ (say).

(e) It can be shown that $z = (\hat{p}_1 - p_1 - \hat{p}_2 + p_2)/\sigma$ converges to $N(0, 1)$ in distribution as $n \to \infty$. Use z to construct a hypothesis test for testing the null hypothesis $p_i = 1/3$, $i = 1, 2, 3$ at level α. Give the test statistic and the rejection region in terms of the standard normal cdf.

Solution 2

Let $p_3 = 1 - p_1 - p_2$.
(a) Since $X_i \sim \text{Bin}(n, p_i)$, we have $\mathbb{E}(X_i) = np_i$ and $\text{Var}(X_i) = np_i(1 - p_i)$. Thus

$$\mathbb{E}(\hat{p}_i) = \frac{\mathbb{E}(X_i)}{n} = p_i, \qquad \text{Var}(\hat{p}_i) = \frac{\text{Var}(X_i)}{n^2} = \frac{p_i(1 - p_i)}{n}, \quad i = 1, 2.$$

(b) The third count is determined by the first two, $X_3 = n - X_1 - X_2$. Thus the joint pmf

of (X_1, X_2) is

$$P(X_1 = x_1, X_2 = x_2) = \frac{n!}{x_1!\, x_2!\, (n - x_1 - x_2)!}\, p_1^{x_1} p_2^{x_2} (1 - p_1 - p_2)^{n - x_1 - x_2},$$

for integers $x_1, x_2 \geq 0$ with $x_1 + x_2 \leq n$.

(c) We compute $\mathbb{E}(X_1 X_2)$ from the joint pmf. The terms with $x_1 = 0$ or $x_2 = 0$ vanish. For $x_1, x_2 \geq 1$, we cancel x_1 and x_2 with the factorials to get

$$\begin{aligned}
\mathbb{E}(X_1 X_2) &= \sum_{x_1, x_2 \geq 1} \frac{n!}{(x_1 - 1)!\, (x_2 - 1)!\, (n - x_1 - x_2)!}\, p_1^{x_1} p_2^{x_2} p_3^{n - x_1 - x_2} \\
&= n(n-1) p_1 p_2 \sum_{x_1, x_2 \geq 1} \frac{(n-2)!}{(x_1 - 1)!\, (x_2 - 1)!\, (n - x_1 - x_2)!}\, p_1^{x_1 - 1} p_2^{x_2 - 1} p_3^{n - x_1 - x_2} \\
&= n(n-1) p_1 p_2\, (p_1 + p_2 + p_3)^{n-2} = n(n-1) p_1 p_2,
\end{aligned}$$

since the last sum is the total mass of a $\text{Multinomial}_3(n-2, (p_1, p_2, p_3))$ distribution. Thus

$$\text{Cov}(X_1, X_2) = \mathbb{E}(X_1 X_2) - \mathbb{E}(X_1)\mathbb{E}(X_2) = n(n-1) p_1 p_2 - n^2 p_1 p_2 = -n p_1 p_2,$$

and so

$$\text{Cov}(\hat{p}_1, \hat{p}_2) = \frac{\text{Cov}(X_1, X_2)}{n^2} = -\frac{p_1 p_2}{n}.$$

(d) We have

$$\begin{aligned}
\sigma^2 = \text{Var}(\hat{p}_1 - \hat{p}_2) &= \text{Var}(\hat{p}_1) + \text{Var}(\hat{p}_2) - 2\,\text{Cov}(\hat{p}_1, \hat{p}_2) \\
&= \frac{p_1(1 - p_1)}{n} + \frac{p_2(1 - p_2)}{n} + \frac{2 p_1 p_2}{n} \\
&= \frac{p_1 + p_2 - (p_1 - p_2)^2}{n}.
\end{aligned}$$

(e) Under $H_0 : p_1 = p_2 = p_3 = 1/3$ we have $p_1 - p_2 = 0$, so that $\hat{p}_1 - p_1 - \hat{p}_2 + p_2 = \hat{p}_1 - \hat{p}_2$ and

$$\sigma_0^2 = \frac{1/3 + 1/3 - 0}{n} = \frac{2}{3n}.$$

Thus, the test statistic is

$$z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{2/(3n)}} = \frac{X_1 - X_2}{\sqrt{2n/3}} \xrightarrow{d} N(0, 1) \quad \text{under } H_0.$$

Large values of $|z_0|$ indicate departure from the null. Let Φ be the standard normal cdf. The level α test (approximately, for large n) rejects H_0 if

$$|z_0| > \Phi^{-1}(1 - \tfrac{\alpha}{2}),$$

i.e., the rejection region is $\{z_0 < -\Phi^{-1}(1 - \alpha/2)\} \cup \{z_0 > \Phi^{-1}(1 - \alpha/2)\}$. The p-value is $2\{1 - \Phi(|z_0|)\}$.

Remark. This statistic only uses $\hat{p}_1 - \hat{p}_2$, so it can detect only those departures from H_0 for which $p_1 \neq p_2$. It does not have power against alternatives like $p_1 = p_2 \neq p_3$.

Question 3

The tensile strength of Portland cement is being studied. Four different mixing techniques can be used economically. A completely randomized experiment was conducted and the following data was collected.

Technique	Tensile strength (lb/in^2)			
1	3129	3000	2865	2890
2	3200	3300	2975	3150
3	2800	2900	2985	3050
4	2600	2700	2600	2765

(a) Test the hypothesis that mixing techniques affect the tensile strength of the cement. Use $\alpha = 0.05$.

(b) Construct a normal probability plot of residuals. What conclusion can be drawn?

(c) Plot the residuals vs the fitted values. What conclusion can be drawn?

(d) Make comparisons between pairs of means without any adjustment for multiple comparisons.

(e) Use Bonferroni method to make comparisons between pairs of means.

(f) Do the two methods give different results? If so, explain where this difference is coming from.

Solution 3

Here $k = 4$, $n_i = 4$ for each technique and $N = 16$. We fit the one-way model $y_{ij} = \mu_i + \varepsilon_{ij}$, $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$.

Listing 1: One-way ANOVA for the cement data

```
y    <- c(3129, 3000, 2865, 2890, 3200, 3300, 2975, 3150,
          2800, 2900, 2985, 3050, 2600, 2700, 2600, 2765)
tech <- factor(rep(1:4, each = 4))

fit <- aov(y ~ tech)
summary(fit)
tapply(y, tech, mean)
```

(a) The level means are $\bar{y}_{1\cdot} = 2971.00$, $\bar{y}_{2\cdot} = 3156.25$, $\bar{y}_{3\cdot} = 2933.75$, $\bar{y}_{4\cdot} = 2666.25$ and the grand mean is $\bar{y}_{\cdot\cdot} = 2931.81$. We want to test $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ against H_1 : not all μ_i are equal. The sums of squares are

$$SS_{Tr} = \sum_{i=1}^{4} 4(\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2 = 489740.19, \qquad SS_E = \sum_{i=1}^{4}\sum_{j=1}^{4}(y_{ij} - \bar{y}_{i\cdot})^2 = 153908.25.$$

Source	df	SS	MS	F	p-value
Technique	3	489740.19	163246.73	12.73	0.00049
Error	12	153908.25	12825.69		
Total	15	643648.44			

Table 1: ANOVA table for the cement data.

Under H_0, $F \sim F_{3,12}$ and the critical value is $F_{3,12;\,0.05} = 3.49$. Since $12.73 > 3.49$ (equivalently $p = 0.00049 < 0.05$) we reject H_0. Thus, the mixing technique has a significant effect on the tensile strength. Also, $\hat{\sigma}^2 = MSE = 12825.69$ (so $\hat{\sigma} = 113.25$) and $R^2 = 489740.19/643648.44 = 0.761$.

Listing 2: Residual diagnostics

```r
r <- resid(fit)
qqnorm(r); qqline(r)
plot(fitted(fit), r, xlab = "Fitted", ylab = "Residuals")
abline(h = 0, lty = 2)
shapiro.test(r)
```

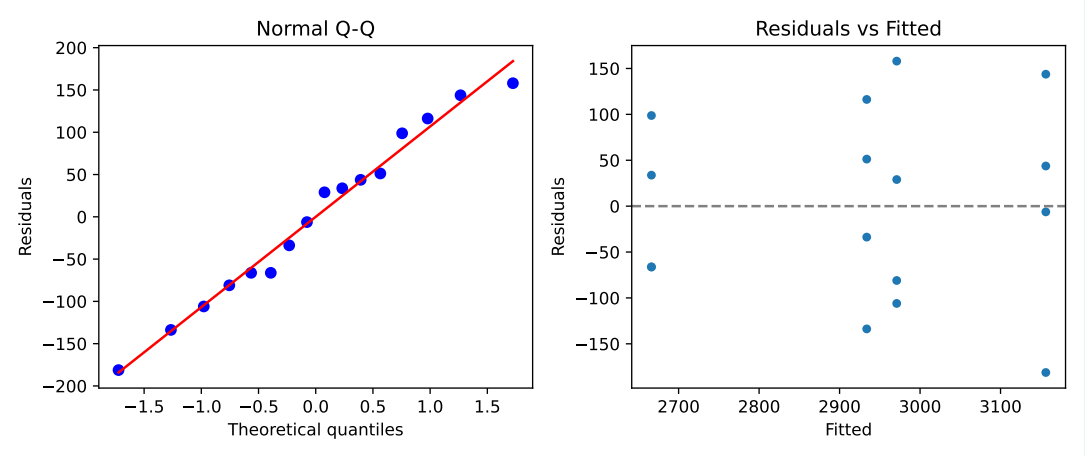


Figure 1: Normal probability plot and residuals vs fitted values for the cement data.

(b) In the normal probability plot the points lie close to the reference line with no systematic curvature or heavy tails. The Shapiro–Wilk test also does not reject normality ($W = 0.970$, $p = 0.846$). Thus, the normality assumption of the errors looks reasonable.
(c) The residuals are scattered around zero with no funnel shape or curvature, and the spread is similar across the four fitted values (the group means). Thus, the constant variance assumption looks reasonable. (With only four observations per group, this is a rough check.)
(d) Without any adjustment, we compare each pair using the least significant difference (LSD)

$$\text{LSD} = t_{12;\,0.025}\sqrt{MSE\left(\tfrac{1}{4}+\tfrac{1}{4}\right)} = 2.179 \times \sqrt{6412.84} = 174.48.$$

We declare $\mu_i \neq \mu_j$ if $|\bar{y}_{i\cdot} - \bar{y}_{j\cdot}| > 174.48$. The standard error of each difference is $\sqrt{6412.84} = 80.08$.

Listing 3: Pairwise comparisons

```r
pairwise.t.test(y, tech, p.adjust.method = "none")
pairwise.t.test(y, tech, p.adjust.method = "bonferroni")
```

Pair	$\lvert\bar{y}_i - \bar{y}_j\rvert$	t	p (unadj.)	p (Bonf.)	LSD	Bonferroni
1 vs 2	185.25	−2.313	0.0392	0.2354	Significant	Not significant
1 vs 3	37.25	0.465	0.6502	1.0000	Not significant	Not significant
1 vs 4	304.75	3.806	0.0025	0.0150	Significant	Significant
2 vs 3	222.50	2.778	0.0167	0.1002	Significant	Not significant
2 vs 4	490.00	6.119	0.00005	0.0003	Significant	Significant
3 vs 4	267.50	3.340	0.0059	0.0353	Significant	Significant

Table 2: All six pairwise comparisons at $\alpha = 0.05$.

Without adjustment, five of the six pairs are significantly different. Only techniques 1 and 3 cannot be distinguished.

(e) There are $m = \binom{4}{2} = 6$ comparisons. The Bonferroni method tests each pair at level $\alpha/m = 0.05/6 \approx 0.0083$, i.e., it declares $\mu_i \neq \mu_j$ if

$$|\bar{y}_{i\cdot} - \bar{y}_{j\cdot}| > t_{12;\,0.05/12}\sqrt{MSE\left(\tfrac{1}{4} + \tfrac{1}{4}\right)} = 3.153 \times 80.08 = 252.47.$$

Equivalently, the Bonferroni adjusted p-value is $\min(1, 6p)$ (see the table above). Only three pairs are now significant: (1,4), (2,4) and (3,4). Thus, technique 4 has a significantly lower mean strength than every other technique, while techniques 1, 2 and 3 cannot be separated from each other.

(f) Yes, the two methods differ for the pairs (1,2) and (2,3). They are declared different by the unadjusted method (differences 185.25 and 222.50 exceed 174.48) but not by Bonferroni (both are below 252.47). The reason is the multiple comparison problem. Each unadjusted test has a type I error probability of 0.05, but we are performing 6 tests simultaneously. If the tests were independent, then the probability of at least one false rejection under the global null would be $1 - 0.95^6 \approx 0.26$. The tests are not independent, but the family-wise error rate is still well above 0.05. Bonferroni controls the family-wise error rate at $\alpha = 0.05$ by using the more stringent per-test level $\alpha/6$, which makes the critical difference larger (252.47 vs 174.48). This protection comes at the cost of power, so the Bonferroni method is more conservative.

Question 4

Consider an experiment with continuous response (y) and one factor (x) with k levels. Suppose there are n_i independent measurements at each level for $i = 1, \ldots, k$. A linear combination $\sum c_i\beta_i$ of the level means (β_i) is called a contrast if $\sum c_i = 0$. For e.g., $\beta_2 - \beta_1$ is a contrast.

(a) Write down the linear model and explain the problem of non-identifiability.

(b) If we want to obtain point estimates of the level means, then what should be the procedure for removing non-identifiability?

(c) Suppose we estimate a contrast by plugging in the point estimates of each level mean ($\sum_{j=1}^{n_i} y_{ij}/n_i$). Find the variance of this estimate.

(d) Using the above and assuming independent normal errors, construct a test for the hypothesis $H_0 : \sum c_i\beta_i = 0$.

Solution 4

Let $N = \sum_{i=1}^{k} n_i$.
(a) Consider the following model

$$y_{ij} = \mu + \alpha_i + \varepsilon_{ij}, \quad i = 1, \dots, k,\ j = 1, \dots, n_i$$

with $\mathbb{E}(\varepsilon_{ij}) = 0$ and $\mathrm{Var}(\varepsilon_{ij}) = \sigma^2$. In matrix form $Y = X\beta + \varepsilon$ where $\beta = (\mu, \alpha_1, \dots, \alpha_k)^T$ and X is $N \times (k+1)$. The first column of X (intercept) is the sum of the remaining k indicator columns, so $\mathrm{rank}(X) = k < k+1$ and X^TX is singular. Equivalently, for any constant c, the parameters $(\mu + c, \alpha_1 - c, \dots, \alpha_k - c)$ give the same $\mathbb{E}(Y)$. Thus, β is not identifiable and the normal equations have infinitely many solutions. Only the level means $\beta_i = \mu + \alpha_i$ are identifiable.
(b) Since we only want the level means, we remove the intercept (equivalently, impose the constraint $\mu = 0$) and fit the cell means model

$$y_{ij} = \beta_i + \varepsilon_{ij}.$$

The design matrix is now $N \times k$ with columns $\mathbf{1}_{n_1}, \dots, \mathbf{1}_{n_k}$ (placed in the blocks of the respective levels), which has full column rank k. Then $X^TX = \mathrm{diag}(n_1, \dots, n_k)$ and

$$\hat{\beta} = (X^TX)^{-1}X^TY = (\bar{y}_{1\cdot}, \dots, \bar{y}_{k\cdot})^T, \qquad \mathrm{Var}(\hat{\beta}) = \sigma^2\,\mathrm{diag}\Big(\tfrac{1}{n_1}, \dots, \tfrac{1}{n_k}\Big).$$

(c) The estimate of the contrast $\sum c_i\beta_i$ is $\sum_{i=1}^{k} c_i\bar{y}_{i\cdot}$. Since the $\bar{y}_{i\cdot}$ are uncorrelated (they are based on different observations), we have

$$\mathrm{Var}\left(\sum_{i=1}^{k} c_i\bar{y}_{i\cdot}\right) = \sum_{i=1}^{k} c_i^2\,\mathrm{Var}(\bar{y}_{i\cdot}) = \sigma^2\sum_{i=1}^{k}\frac{c_i^2}{n_i}.$$

For example, for $\beta_2 - \beta_1$ this is $\sigma^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)$.
(d) Assuming $\varepsilon_{ij} \overset{\text{iid}}{\sim} N(0, \sigma^2)$, we have

$$\sum c_i\bar{y}_{i\cdot} \sim N\left(\sum c_i\beta_i,\ \sigma^2\sum\frac{c_i^2}{n_i}\right).$$

The unbiased estimator of σ^2 is $\hat{\sigma}^2 = MSE = \frac{1}{N-k}\sum_i\sum_j(y_{ij} - \bar{y}_{i\cdot})^2$ and $(N-k)\hat{\sigma}^2/\sigma^2 \sim \chi^2_{N-k}$. Also $\hat{\sigma}^2$ is independent of $\hat{\beta}$. Thus, under $H_0 : \sum c_i\beta_i = 0$ the test statistic

$$t = \frac{\sum_{i=1}^{k} c_i\bar{y}_{i\cdot}}{\sqrt{MSE\sum_{i=1}^{k} c_i^2/n_i}} \ \sim\ t_{N-k}.$$

We reject H_0 against $H_1 : \sum c_i\beta_i \neq 0$ at level α if $|t| > t_{N-k;\,\alpha/2}$. The p-value is $2P(t_{N-k} > |t|)$. Equivalently, we reject when $t^2 > F_{1,N-k;\,\alpha}$, since $t^2 \sim F_{1,N-k}$ under H_0.

Question 5

Consider the Forbes2000 data in the R library HSAUR.

(a) For the variable category draw a bar chart.

(b) For the variable country draw a bar chart.

(c) Take the four categories that have the highest frequency and club every other category as others. Take the four countries that have the highest frequency and club every other country as others. Restrict your dataset to combinations of these (25 combinations).

(d) Draw a stacked bar plot with these 25 combinations and interpret the results.

Solution 5

The data has 2000 companies with 27 categories and 61 countries.

Listing 4: Bar charts for category and country

```r
library(HSAUR)
data(Forbes2000)

cat_tab <- sort(table(Forbes2000$category), decreasing = TRUE)
cty_tab <- sort(table(Forbes2000$country),  decreasing = TRUE)

par(mar = c(5, 12, 3, 1))
barplot(rev(cat_tab), horiz = TRUE, las = 1, xlab = "Frequency")
barplot(rev(cty_tab), horiz = TRUE, las = 1, xlab = "Frequency", cex.
    names = 0.6)
```

(a) The bar chart of category is given in Figure 2. Banking is by far the most frequent category (313 companies), followed by Diversified financials (158), Insurance (112) and Utilities (110). Materials (97) is the next one.

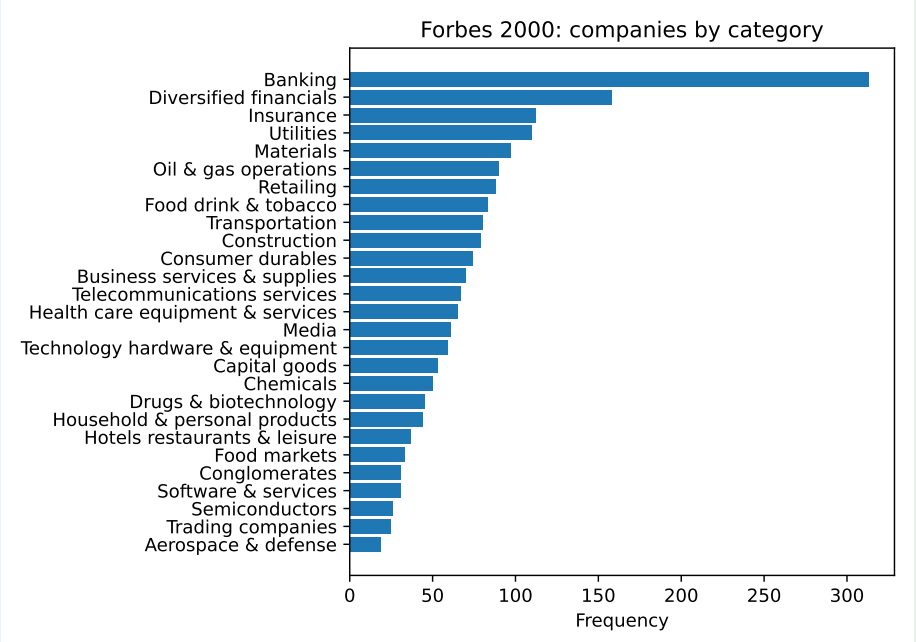


Figure 2: Bar chart of category.

(b) The bar chart of country is given in Figure 3. The distribution is very skewed: the United States (751) and Japan (316) together account for more than half of the companies. They are followed by the United Kingdom (137), Germany (65) and France (63), and many countries have only a handful of companies.

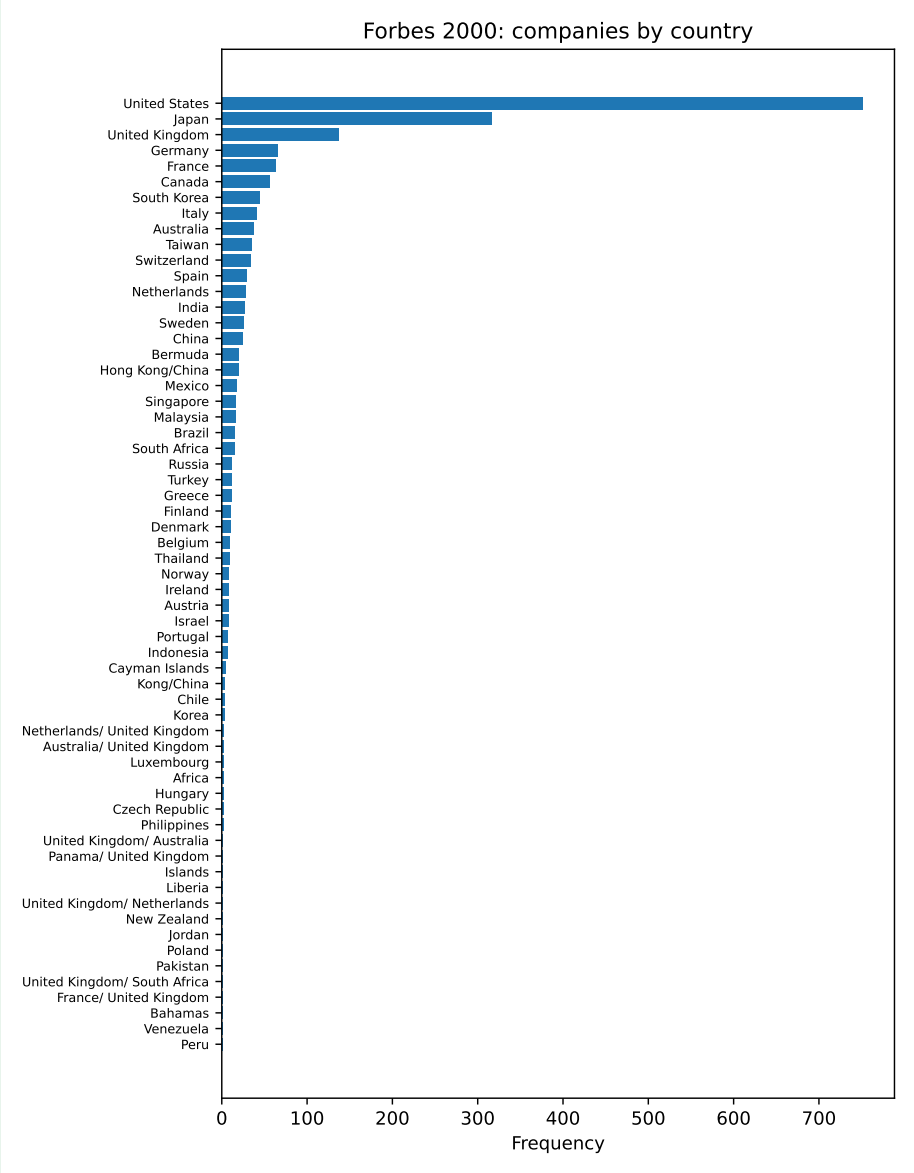


Figure 3: Bar chart of country.

(c) The four most frequent categories are Banking, Diversified financials, Insurance and Utilities and the four most frequent countries are United States, Japan, United Kingdom and Germany. There are no ties at the cut-off (Utilities 110 vs. Materials 97; Germany 65 vs. France 63). We club all other categories and countries as "Others", which gives $5 \times 5 = 25$ combinations.

Listing 5: Clubbing and the 25 combinations

```
top_cat <- names(cat_tab)[1:4]
top_cty <- names(cty_tab)[1:4]
```

```r

cat2 <- ifelse(Forbes2000$category %in% top_cat, as.character(Forbes2000$category), "Others")
cty2 <- ifelse(Forbes2000$country  %in% top_cty, as.character(Forbes2000$country),  "Others")

tab <- table(Country = factor(cty2, levels = c(top_cty, "Others")),
             Category = factor(cat2, levels = c(top_cat, "Others")))
tab
addmargins(tab)
```

	Banking	Div. fin.	Insurance	Utilities	Others	Total
United States	83	60	46	54	508	751
Japan	69	24	9	11	203	316
United Kingdom	9	13	8	10	97	137
Germany	6	8	4	4	43	65
Others	146	53	45	31	456	731
Total	313	158	112	110	1307	2000

Table 3: Frequencies of the 25 combinations of country and category.

(d) We draw the stacked bar plot with one bar per country (the segments are the categories).

Listing 6: Stacked bar plot

```r
cols <- c("#1f77b4", "#ff7f0e", "#2ca02c", "#d62728", "#9467bd")
barplot(t(tab), col = cols, xlab = "Country", ylab = "Frequency",
        legend.text = colnames(tab), args.legend = list(title = "Category"))
```

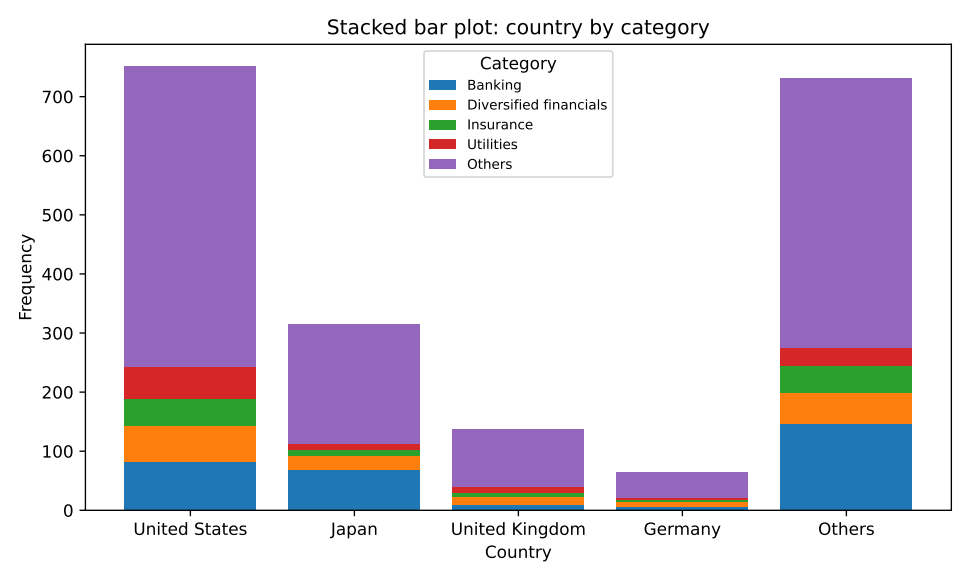


Figure 4: Stacked bar plot of country by category.

To compare the countries fairly, we look at the row percentages (proportion of the companies of a country in each category).

	Banking	Div. fin.	Insurance	Utilities	Others
United States	11.1	8.0	6.1	7.2	67.6
Japan	21.8	7.6	2.8	3.5	64.2
United Kingdom	6.6	9.5	5.8	7.3	70.8
Germany	9.2	12.3	6.2	6.2	66.2
Others	20.0	7.3	6.2	4.2	62.4

Table 4: Percentage of companies of each country in each category.

We observe the following:

- The United States has the tallest bar (751). Among the four named countries it has the most companies in every category, e.g., 83 banks against 69 for Japan.
- The bars of the United Kingdom and Germany are short, and these two countries have very few banks (9 and 6).
- Japan has a large share in banking: 21.8% of its companies are banks compared to 11.1% for the US and only 6.6% for the UK. On the other hand, Japan has relatively few Insurance and Utilities companies (2.8% and 3.5%).
- The "Others" category dominates every bar (62–71%), so the four chosen categories together make up only about one-third of the companies in each country.
- The distributions are not identical across countries. A Pearson chi-square test of independence on the 5×5 table gives $\chi^2 = 54.6$ on 16 df ($p \approx 4 \times 10^{-6}$). Some expected counts are small (e.g., about 3.6 for Germany and Insurance), so this should be taken as indicative only.

Thus, the country and the industry category are associated, with the clearest feature being Japan's high banking share.